

Evolutionary timescale (deep time)

1

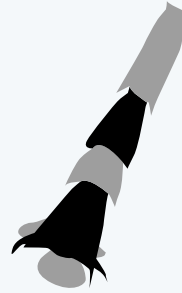
Evolutionary conservation

Three conserved sensory appendages with distinct anatomical and functional identities appear to have stabilized early in brachyceran evolution.

Antenna
(olfaction)

Maxillary palp
(close-range
chemoreception)

Tarsi
(contact, thermo-
mechano-sensation)



Key idea

Appendage identities represent sensory modules that are evolutionary conserved across Diptera.

Evidence from *H. illucens*

Appendage identity explains ~91% of chemosensory transcriptomic variance.

Distinct, coherent expression profiles for antenna, palps and tarsi.

Evolutionary timescale (intermediate time)

2

Molecular diversification

Within conserved appendage architectures, chemosensory gene families expanded, contracted and specialized to support ecological adaptations.

ORs

GRs

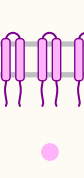
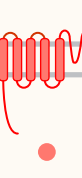
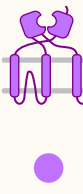
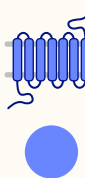
IRs

OBP

CSPs

TRPs

PPKs



Key idea

Sensory innovation occurs primarily through diversification and differential deployment of receptor families within conserved appendages rather than through changes in appendage identity.

Evidence from *H. illucens*

Extensive lineage-specific expansions of ORs and TRPs.

Conserved CO₂ receptor-like GRs divided between maxillary palps and antennae, presenting both dipteran arrangements.

Adult timescale (short term)

3

Physiological modulation fine-tunes conserved sensory circuits

Sex, reproductive state and circadian context adjust the activity and sensitivity of existing sensory circuits through modulation of conserved regulatory pathways rather than remodeling receptor repertoires.

Sex

Reproductive state

Circadian context

Key idea

Adult behavioral transitions are achieved by changing the operating state of conserved sensory modules through regulation of shared signaling pathways.

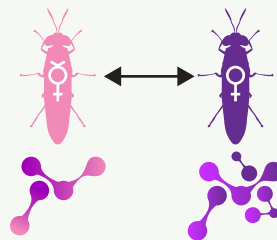
Evidence from *H. illucens*

Sex explains ~1.4% and mating status ~1% of transcriptomic variance.

Consistent regulation of genes in sex determination, circadian, neuropeptide and extracellular pathways.



Sex determination
(*tra*)



Neuropeptide signaling
(SIFamide receptor)
Extracellular remodeling
Metabolism



Circadian regulation
(Daywake)

Integrated outcome

Stable appendage-specific architectures, diversified receptor repertoires, regulatory plasticity support flexible, efficient chemosensory systems that support diverse behaviors